

Operators and Control Statements & Looping

**SIMATS**
ENGINEERING

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JAVA PROGRAMMING

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QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR Java Run Save**OUTPUT****INPUT**

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MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR Java Run Save**OUTPUT****INPUT**



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Leap

LEAP
Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class leap
3 {
4     public static void main(String[] args)
5     {
6         Scanner reader =new Scanner(System.in);
7         int year =reader.nextInt();
8         if( year%4==0 && year%100!=0 || year%400==0)
9         {
10             System.out.println(year + " is a leap year.");
11         }
12         else
13         {
14             System.out.println(year + " is a not leap year.");
15         }
16     }
17 }
18 }
```

INPUT
2020

OUTPUT
2020 is a leap year.



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Fibonacci

FIBONACCI
Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Fibonacci
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int limit = sc.nextInt();
8         int firstTerm = 0;
9         int SecondTerm = 1;
10        System.out.println("Fibonacci up to " +limit + ":");
11        while (firstTerm <= limit) {
12            System.out.println(firstTerm + " ");
13            int nextTerm = firstTerm + SecondTerm;
14            firstTerm = SecondTerm;
15            SecondTerm= nextTerm;
16        }
17    }
18 }
19 }
```

INPUT
4
5
6

OUTPUT
Fibonacci up to 4:
0
1
1
2



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QUESTIONS 31 / 31 completed

Factorial

FACTORIAL
Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 public class Second
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc= new Scanner(System.in);
7         int num ;
8         num = sc.nextInt();
9         int fact = 1;
10        for (int i = 2; i<=num; i++)
11        {
12            fact = fact * i;
13        }
14        System.out.println("factorial = " + fact);
15    }
16 }
17 }
18 }
19 }
```

INPUT
3

OUTPUT
factorial = 6



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QUESTIONS 2/7 3/5 completed

RT-Pattern

RT-PATTERN
Write a Java program to print the following pattern using loops:
**


```
1 import java.util.Scanner;
2 public class Starpattern
3 {
4     public static void main(String[] args)
5     {
6         for(int i = 0; i<=5; i++)
7         {
8             for(int j = 0; j<=i; j++)
9             {
10                 System.out.print("**");
11             }
12             System.out.println();
13         }
14     }
15 }
16 }
```

OUTPUT

**

4



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QUESTIONS 2/7 3/5 completed

GCD

GCD
Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class GCD
3 {
4     public static void main(String[] args)
5     {
6         Scanner Scanner = new Scanner(System.in);
7         System.out.print("enter first number: ");
8         int num1 = Scanner.nextInt();
9         System.out.print("enter second number: ");
10        int num2 = Scanner.nextInt();
11        int n1=num1;
12        int n2= num2;
13        while (num1!=num2)
14        {
15            if (num1>num2)
16            {
17                num1 = num1 - num2;
18            }
19            else
20            {
21                num2 = num2 - num1;
22            }
23        }
24    }
25 }
```

OUTPUT

enter first number: enter second number: The GCD of 2 and 4is: 2

2
4



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QUESTIONS 2/7 3/5 completed

Prime

PRIME
Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class Prime
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc= new Scanner(System.in);
7         int a = sc.nextInt();
8         int count = 0;
9         for (int i = 1; i<=a; i++)
10        {
11            if (a%i==0)
12            {
13                count++;
14            }
15        }
16        if(count==2)
17        {
18            System.out.println("prime number");
19        }
20        else
21        {
22            System.out.println("composite number");
23        }
24    }
25 }
```

OUTPUT

composite number

10



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Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

```
1 import java.util.Scanner;
2 public class Multiplicationtable
3 {
4     public static void main(String[] args)
5     {
6         Scanner Scanner = new Scanner(System.in);
7         System.out.print("enter a number: ");
8         int number = Scanner.nextInt();
9         System.out.print("Multiplication Table for " + number + " :");
10        for(int i = 1; i<=10; i++)
11        {
12            System.out.println(number + " x " + i + " = " +(number * i));
13        }
14    }
15 }
16 }
```

OUTPUT

enter a number: Multiplication Table for 5 :5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30

5



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QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class Starpattern
3 {
4     public static void main(String[] args)
5     {
6         for(int i=0; i<=5; i++){
7             for(int j=i; j<=5; j++){
8                 System.out.print("*");
9             }
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

**

5



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QUESTIONS 31 / 31 completed

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR
Java Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9
10        int lcm = (a > b) ? a : b;
11
12        while (true) {
13            if (lcm % a == 0 && lcm % b == 0) {
14                System.out.print(lcm);
15                break;
16            }
17            lcm++;
18        }
19    }
20 }
```

OUTPUT

45

INPUT

5
9



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D to B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR
Java Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int decimal = 11;
4         int binary = 0, place = 1;
5
6         while (decimal > 0) {
7             int rem = decimal % 2;
8             binary += rem * place;
9             place *= 10;
10            decimal /= 2;
11        }
12
13        System.out.println(binary);
14    }
15 }
```

OUTPUT

1011

INPUT

Your INPUT goes here!



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QUESTIONS 31 / 31 completed

B to D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR
Java Run Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int binary = 1011;
4         int decimal = 0, base = 1;
5
6         while (binary > 0) {
7             int rem = binary % 10;
8             decimal += rem * base;
9             base *= 2;
10            binary /= 10;
11        }
12
13        System.out.println(decimal);
14    }
15 }
```

OUTPUT

11

INPUT

Your INPUT goes here!



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QUESTIONS 31 / 31 completed

Sum

SUM
Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class sumDigits
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int sum = 0;
9         while (n!= 0)
10        {
11            sum +=n%10;
12            n/=10;
13        }
14        System.out.println("sum of Digits =" +sum);
15    }
16 }
```

OUTPUT

sum of Digits =1

INPUT

12345



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QUESTIONS 31 / 31 completed

Product

PRODUCT
Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int num = 1234; // Change the number here
4         int product = 1;
5
6         while (num > 0) {
7             int digit = num % 10;
8             product = product * digit;
9             num = num / 10;
10        }
11
12        System.out.println(product);
13    }
14 }
```

OUTPUT

24

INPUT

Your INPUT goes here!



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Power

POWER
Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int p = sc.nextInt();
7         int ans = 1;
8         for (int i = 1; i <= p; i++)
9             ans *= n;
10        System.out.println(ans);
11    }
12 }
```

OUTPUT

32

INPUT

2
5



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QUESTIONS 31 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

```
1 import java.util.Scanner;
2 public class sphere
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int r = sc.nextInt();
8         double vloume = (4 / 3) * (3.14)*(r*r*r);
9         System.out.println("vloume of sphere= " +vloume);
10    }
11 }
12 }
```

OUTPUT

vloume of sphere= 392.5

INPUT

5



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Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class palindrome
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n=sc.nextInt();
8         int rev=0;
9         int original = n;
10        while(n!=0)
11        {
12            rev = rev*10+n%10;
13            n/=10;
14        }
15        if (original==rev)
16        {
17            System.out.println("Reverse the Digits");
18            System.out.println("Palindrome");
19        }
20    }
21 }
```

OUTPUT

Reverse the Digits
Palindrome

INPUT

12321



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Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
1 class Main {
2     public static void main(String[] args) {
3         int n = 153; // Input
4         int temp = n, sum = 0;
5
6         while (temp > 0) {
7             int d = temp % 10;
8             sum += d * d * d;
9             temp /= 10;
10        }
11
12        if (sum == n)
13            System.out.println("Armstrong Number");
14        else
15            System.out.println("Not Armstrong Number");
16    }
17 }
```

OUTPUT

Armatrong Number

INPUT

153



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QUESTIONS 31 / 31 completed

Perfect

PERFECT
Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int sum = 0;
9
10        for (int i = 1; i < n; i++) {
11            if (n % i == 0) {
12                sum += i;
13            }
14        }
15
16        if (sum == n) {
17            System.out.println("Perfect Number");
18        } else {
19            System.out.println("Not Perfect Number");
20        }
21    }
22 }
```

OUTPUT
Perfect Number

INPUT
28



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QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY
Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class rectangle
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         System.out.print("enter rows:");
8         int rows=sc.nextInt();
9         for(int i = 1; i <=rows;i++)
10        {
11            for(int j=1; j<=i; j++)
12            {
13                System.out.print(" ");
14            }
15            System.out.println();
16        }
17    }
18 }
19 }
```

OUTPUT
enter rows:
+
+
+
+

INPUT
5



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QUESTIONS 31 / 31 completed

Min

MIN
Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 class Main {
2     public static void main(String[] args) {
3         int[] a = {12, 5, 8, 3, 10};
4
5         int min = a[0];
6
7         for (int i = 1; i < a.length; i++) {
8             if (a[i] < min)
9                 min = a[i];
10        }
11
12        System.out.println(min);
13    }
14 }
```

OUTPUT
3

INPUT
Your INPUT goes here!

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 class Main {
2     public static void main(String[] args) {
3         int[] a = {5, 2, 9, 1, 3};
4
5         for (int i = 1; i < a.length; i++) {
6             int key = a[i];
7             int j = i - 1;
8
9             while (j >= 0 && a[j] > key) {
10                 a[j + 1] = a[j];
11                 j--;
12             }
13             a[j + 1] = key;
14         }
15
16         for (int i = 0; i < a.length; i++)
17             System.out.print(a[i] + " ");
18     }
19 }
```

OUTPUT

1 2 3 5 9

INPUT

Your INPUT goes
here!